

DMBI IA 2 QB

2M Questions

- Q1.** What is clustering?
- Q2.** Explain the different distance measures that can be used to compute distance between two clusters?
- Q3.** Difference between classification and clustering
- Q4.** Types of Clustering Methods?
- Q5.** Explain K-means clustering Algorithm
- Q6.** Explain K-medoids Clustering?
- Q7.** What is hierarchical clustering? Explain any 2 techniques for finding distance.
- Q8.** Explain Birch algorithm with an example.
- Q9.** Explain DB Scan clustering algorithm with an example
- Q10.** What is an outliers? Describe methods that are used for outliers detection analysis.
- Q11.** Explain Market Basket Analysis with an example.
- Q12.** Explain frequent item sets, close item sets and large item sets.
- Q13.** Define and explain support and confidence.
- Q14.** What is apriori algorithm.
- Q15.** Explain Multilevel association rule.
- Q16.** Explain Multidimensional Association rule.
- Q17.** Define Business Intelligence system with example.
- Q18.** Explain Business Intelligence Issues.
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5M Questions

Q19. Use K means algorithm to create 3 clusters for given set of values.
(2, 3, 6, 8, 9, 12, 15, 18, 22) .

Q20. Use K means to cluster the following data sets into 3 clusters.

Protein	20	21	15	22	20	25	26	20	18	20
Fat	9	9	7	17	8	12	14	9	9	9

Q21. Suppose we have six objects with name A, B, C, D, E, F apply single linkage clustering and draw a dendrogram for the given data.

Object	X	Y
A	1	1
B	1.5	1.5
C	5	5
D	3	4
E	4	4
F	3	3.5

Q22. Consider the transaction database given in the table below. Apply **Apriori** algorithm with minimum support of 50% and confidence of 50%. Find all Frequent itemsets and all the association rules.

Tid	Items
100	1,3,4
200	2,3,5
300	1,2,3,5
400	2,5

Q23. Use the **Apriori** algorithm to identify the frequent item-sets in the following database. Then extract the strong association rules from these sets.

Minimum support - 30%, Minimum confidence = 75%.

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TID	ITEMS
01	A, B, D, E, P
02	B, C, E
03	A, B, D, E
04	A, B, C, E
05	A, B, C, D, E, F
06	B, C, D
07	A, B, D, E

Q24. What are multidimensional association rules explain with suitable examples.

Q26. List and explain definition of decision support system with a diagram.

Q27. Explain development of BI system with architecture.